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Capturing differential diffusion effects in large eddy simulation of turbulent premixed flames

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ABSTRACT

The combustion of hydrogen in low-swirl burners (LSB) is considered as an alternative means of generating power because it is characterized by low emissions and high efficiency. However, lean hydrogen premixed flames are subject to thermodiffusive instabilities induced by the large diffusivity, and hence small Lewis number, of hydrogen. The numerical modelling of these flows remains challenging because the transition of small scale instabilities into large scale turbulent structures cannot be modelled by conventional strategies. Recently, Schlup and Blanquart (2019) developed a two-equation model which captures successfully the phenomena arising from differential diffusion and curvature effects. The chemistry tabulation framework is based on the classical progress variable approach and introduces an additional transport equation to account for fluctuations in the local equivalence ratio due to these effects. In the current work, this model is extended to large eddy simulation (LES) of an LSB. The LES model is applied first to a CH_4/air flame ($\phi = 0.59$) to validate the overall simulation framework and then to a H_2/air flame ($\phi = 0.4$). The results obtained with this new formulation show significant improvement over the traditional one-equation formulation. The unique flow field exhibited by lean hydrogen is reproduced successfully using the two-equation model.

1. Introduction

The development of lean premixed hydrogen combustors is a key component in developing cleaner energy generation, primarily due to their low emissions and high efficiency [1]. Although the benefits are promising, using hydrogen as the fuel introduces a number of challenges in the design and simulation of these combustors. Specifically, hydrogen has a high reactivity, fast flame speed, and a propensity to autoignite in air. A successful approach to these problems consists of burning hydrogen lean. However, for lean mixtures close to the flammability limit, the resulting combustion is subject to thermodiffusive instabilities which might lead ultimately to unstable combustion, flashback, blow-off, and noise [2].

Several useful strategies for modelling turbulent premixed flames have been developed. For example, in the thickened flame model [3,4], the flame front is artificially thickened and resolved by enhancing the molecular diffusion. However, the enhanced molecular diffusion can damp out critical small scale flame instabilities that might be important in hydrogen combustion. Transported probability density function (PDF) methods are another modelling strategy [5,6]. These methods may be used in conjunction with detailed chemistry, but they remain sensitive to the micro-mixing model used. As a result, it is unclear whether these models can capture hydrogen instabilities properly.

In order to reduce the cost of the numerical simulations and the modelling of the detailed chemical processes, other strategies rely on the use of only a few scalar equations. For instance, methods such as FPI (flame prolongation of ILDM) [7] and FGM (flamelet generated manifolds) [8] tabulate the chemical response of laminar flames with respect to a progress variable. For premixed flames, the simplest method is to create a lookup table based on a single one-dimensional flat unstretched flame [9]. The flame is parameterized by a "progress" variable, C, which defines uniquely the thermochemical state along the combustion trajectory. It is typically a linear combination of the mass fractions of intermediate species and products [10]. Its transport equation takes the form

$$\partial_t(\rho C) + \nabla \cdot (\rho \mathbf{u} C) = \nabla \cdot \left(\rho D_C \nabla C\right) + \dot{\omega}_C \tag{1}$$

Here, ρ is the density, $\mathbf{u}$ is the velocity vector, D_C is a diffusion coefficient, and $\dot{\omega}_C$ is the source term for C. At each timestep, important thermophysical properties are obtained via the lookup table.

Unfortunately, hydrogen exhibits significant differential diffusion effects due to its low Lewis number. This differential diffusion creates pockets of lean or rich mixtures, resulting in local fluctuations of the equivalence ratio and burning velocity. As such, the flame burns

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unevenly, and curvature effects become important, especially in the presence of turbulence. In these cases, the tabulation based on just C is insufficient to describe the physics [9,11].

To account for differential diffusion, several strategies have been proposed. De Swart et al. [12] derived a generic form of the transport equation for any control variable (e.g., progress variable) which incorporates differential diffusion. The model was used by Donini et al. [13] with three controlling variables: a progress variable, enthalpy, and a mixture fraction. Recently, Mukundakumar et al. [14] updated the model to allow for flames which are dominated by differential diffusion effects. The transport equations for the control variables used in these studies include additional source terms due to differential diffusion and chemistry which require tabulation. While these studies generate the tables from one-dimensional flat flames, recent efforts have considered using stretched and curved flames [15,16].

Regele et al. [17] proposed an alternative formulation for non-unity Lewis number fuels based on a two-equation model (progress variable and mixture fraction). This formulation is simpler than that of de Swart et al. [12] since the progress variable equation remains unchanged (no additional source terms), the mixture fraction equation has no chemical source term, and the differential diffusion terms are linear in the transported quantities. Schlup and Blanquart [18] extended the model to a more general formulation including Soret diffusion with full detailed transport. This model was validated in Direct Numerical Simulations (DNS) across a wide range of laminar and turbulent flames [19], including one-dimensional flames (unstretched steady flat and outwardly-propagating cylindrical flames), two-dimensional flames (planar and tubular flames), and finally three-dimensional (laminar and turbulent flames) flames. In all test cases, the model was shown to capture accurately curvature effects and reproduce results from detailed chemistry.

The objectives of the present work are two-fold: (1) to apply the two-equation model of Schlup and Blanquart [18] in the framework of large eddy simulations (LES) and (2) to assess its ability to capture burning behaviour which is not captured through traditional methods in the simulation of a low-swirl burner (LSB).

The paper is organized as follows. First, the governing equations and numerical methods are summarized in Section 2. Then, the simulation configuration and methodology are provided in Section 3. The chemical models and associated LES modelling are elaborated in Section 4. The LES results are presented in Section 5, and the conclusions are presented in Section 6.

2. Governing equations

In the framework of LES, the filtered Navier–Stokes equations are

$$\partial_t \overline{\rho} + \nabla \cdot \left(\overline{\rho}\widetilde{\mathbf{u}}\right) = 0 \tag{2}$$

$$\partial_t (\overline{\rho}\widetilde{\mathbf{u}}) + \nabla \cdot \left(\overline{\rho}\widetilde{\mathbf{u}} \otimes \widetilde{\mathbf{u}}\right) = -\nabla \overline{p} + \nabla \cdot \boldsymbol{\tau} \tag{3}$$

where p is the hydrodynamic pressure, and the turbulent stress tensor terms are expressed using a standard Smagorinsky model

$$\tau = \left(\bar{\mu} + \mu_t\right)\left(\nabla\widetilde{\mathbf{u}} + \nabla\widetilde{\mathbf{u}}^t\right) \tag{4}$$

where μ is the dynamic viscosity. The overline, $\overline{(\,)}$, denotes a filtered quantity, and the tilde denotes a Favre average, $\widetilde{\phi} = \overline{\rho\phi}/\overline{\phi}$. The turbulent eddy viscosity, μ_t, is evaluated using the Lagrangian dynamic subfilter scale model of Meneveau et al. [20]. The transport equations for the filtered scalar fields are discussed later in Section 4.

The present LES are performed with the NGA code [21], which employs fully conservative finite difference schemes for the simulation of variable density low Mach number turbulent flows. Both the spatial and temporal discretizations are second-order accurate, and the time integration is carried out via the semi-implicit iterative Crank–Nicolson method [22]. The scalar fluxes are evaluated using the WENO5 scheme [23].

Two separate fuel/air mixtures are considered. For hydrogen combustion, the 9 species 54 reaction (forward and backward counted separately) chemical mechanism by Hong et al. [24] was used with updated rate constants [25,26]. For methane, the GRI 3.0 mechanism [27] was used. The equivalence ratios under consideration are $\phi = 0.4$ for the hydrogen/air flame (laminar flame speed $S_L = 0.206$ m/s and laminar flame thickness $l_F = 6.8 \times 10^{-4}$ m) and $\phi = 0.59$ for methane/air ($S_L = 0.102$ m/s and $l_F = 1.0 \times 10^{-3}$ m).

3. Burner geometry and boundary conditions

We consider the series of experimental measurements performed by Cheng et al. [1] in an LSB. A brief overview of the geometry is presented here. For a more detailed description, the reader is referred to [1].

As shown in Fig. 1, the low-swirl injector (LSI) is composed of two parts: (1) a central pipe ($R_c = 1.89$ cm) that holds a perforated screen, consisting of 54 holes with a blockage ratio of about 46%, and (2) a co-annular channel ($R_i = 2.86$ cm) composed of 16 vanes with 40° exit blade angles. In the simulations, 42% of the flow passes through the central pipe. After a recess region, the premixed gas mixture of fuel and air is injected into an enclosed axisymmetric combustion chamber with inner radius $R_e = 9$ cm. The combustion chamber has a length of $L = 31$ cm, after which there is a contraction and an exit section.

To simplify the numerical simulations, the computational domain was decomposed into two parts: the swirler and the combustion chamber. The flow through the swirler was investigated first with non-reacting LES, and the outflow conditions downstream of the injector but upstream of the combustion chamber were recorded as a function of time. Then, these outflow conditions were used as inflow conditions for the reacting LES of the combustion chamber. As a result, the flow through the swirler needs to be computed only once, and simulations of the turbulent flame with different fuels can be performed on a coarser mesh. This first calculation is meant to produce realistic turbulence fluctuations with the correct mean velocities. As the kinematic viscosities between the two unburnt mixtures differs by only 11%, a single inflow profile may be generated without significant impact.

Given the complexity of the LSI, the fine details of the geometry of the injector were represented with the Immersed Boundaries (IB) technique [28]. In particular, we use a velocity reconstruction (or interpolation) approach similar to that proposed by Kang et al. [29]. A description of the immersed boundary method is included in the supplementary material. This approach provides an efficient description of the flow around complex geometries on structured meshes. For an accurate description of the flow field, the geometry of the swirler assembly was represented on a cylindrical mesh of $N_x \times N_r \times N_\theta = 256 \times 120 \times 256$ grid points. Fig. 1 shows the contour of velocity magnitude through the swirler. The flow through the central perforated screen resembles grid turbulence and decays slowly with downstream distance. The flow through the swirled vanes transitions rapidly from laminar to turbulent as it is accelerated. The bulk axial velocity through the entire geometry (central pipe and swirled vanes) at the exit plane is $U_0 = 18$ m/s, and the mean azimuthal velocity is 10.2 m/s.

4. Filtered tabulated chemistry approaches

Two separate tabulation strategies are used in the simulations of the turbulent reacting flow in the combustion chamber.

4.1. Model 1: One equation

One-dimensional flat flames serve as an appropriate proxy for higher dimensional flames in which the Lewis numbers, Le, for all species are approximately unity. This is a reasonable assumption for many fuels, including methane. In these cases, the chemistry is tabulated via the progress variable, C. In this work, the progress variable

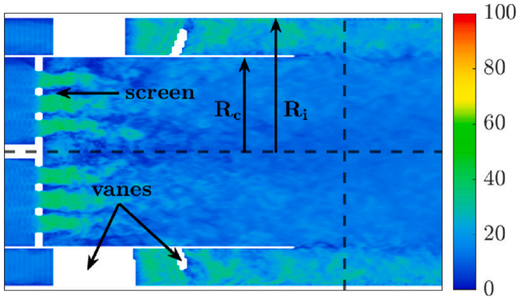


Fig. 1. 2D contour of velocity magnitude through the swirler. The empty white regions correspond to cuts through the walls and swirler vanes. The vertical dashed line represents the location where the inflow conditions for the combustion chamber were extracted.

is defined as the mass fraction of water, Y_{H_2O}. The filtered progress variable transport equation reads

$$\partial_t\left(\bar{\rho}\widetilde{C}\right)+\nabla\cdot\left(\bar{\rho}\widetilde{\mathbf{u}}\widetilde{C}\right)=\nabla\cdot\left(\bar{\rho}\left(\widetilde{D}_C+D_C^t\right)\nabla\widetilde{C}\right)+\bar{\dot{\omega}}_C \tag{5}$$

The eddy diffusivity, D_C^t, is evaluated using the Lagrangian dynamic sub-grid scale model of Meneveau et al. [20]. The eddy diffusivity model is consistent with the formulation of Reveillon and Vervisch [30].

A model must be provided for the filtered source term, $\bar{\dot{\omega}}_C$. In LES, the filtering over a given volume V can be expressed as [31]

$$\bar{\dot{\omega}}_C=\frac{1}{V}\iiint\dot{\omega}_C dV=\int_C\langle\dot{\omega}_C|C\rangle P(C)dC \tag{6}$$

The conditional mean source term, $\langle\dot{\omega}_C|C\rangle$, is taken to be that of the corresponding one-dimensional unstretched flat flame. $P(C)$ represents the subfilter density function, taken to be a presumed β-PDF [32,33]. It is characterized by the mean progress variable, $\widetilde{C}$, and its variance, C_v. The use of a β-PDF has been assessed a priori and a posteriori in low Karlovitz number flames by Mukhopadhyay et al. [34] and high Karlovitz number flames by Lapointe et al. [35,36]. The filtered source term is then expressed as

$$\bar{\dot{\omega}}_C\left(\widetilde{C},C_v\right)=\int_0^{C_{max}}\langle\dot{\omega}_C|C\rangle P\left(C|\widetilde{C},C_v\right)dC \tag{7}$$

The variance of the progress variable is modelled using an algebraic model [37], which takes the form

$$C_v=c_s\Delta^2|\nabla\widetilde{C}|^2 \tag{8}$$

The coefficient c_s is evaluated using the Lagrangian dynamic sub-grid scale model of Meneveau et al. [20], and Δ is the filter width.

4.2. Model 2: Two equations

To account for the effects of differential diffusion, Regele et al. [17] proposed a two-equation model. A mixture fraction-like variable, Z, was included to complement the progress variable and to capture fluctuations in the local equivalence ratio. Its transport equation included source terms resulting from differential diffusion. More recently, Schlup and Blanquart [18] re-derived the Z transport equation by lifting previous limiting assumptions made in [17] and by including Soret diffusion. For details on the derivation of the (C,Z) model, the reader is referred to [18].

The filtered transport equation for $\tilde{Z}$ is given by

$$\begin{aligned}\partial_t\left(\bar{\rho}\widetilde{Z}\right)+\nabla\cdot\left(\bar{\rho}\widetilde{\mathbf{u}}\widetilde{Z}\right)&=\nabla\cdot\left(\bar{\rho}\left(\widetilde{D}_Z+D_Z^t\right)\nabla\widetilde{Z}\right)\\&\quad-\nabla\cdot\left(\bar{\rho}\widetilde{D}_Z^*\nabla\widetilde{C}\right)+\nabla\cdot\left(\bar{\rho}\widetilde{D}_Z^T\nabla\widetilde{T}\right)\end{aligned} \tag{9}$$

where

$$D_Z=\frac{\nu Y_{F,1}D_F+Y_{O,2}D_O}{\nu Y_{F,1}+Y_{O,2}} \tag{10}$$

$$D_Z^*=\left(\frac{1}{\nu+1}\right)\left(\frac{\nu D_F-\nu D_O}{\nu Y_{F,1}+Y_{O,2}}\right) \tag{11}$$

and

$$D_Z^T=\frac{1}{\rho T}\left(\frac{\nu D_F^T-D_O^T}{\nu Y_{F,1}+Y_{O,2}}\right) \tag{12}$$

where D_i and D_i^T represent the molecular and thermodiffusion coefficients, respectively. The term Y indicates the mass fraction, the subscripts O and F indicate oxidizer and fuel, and the subscripts 1 and 2 refer to the fuel and oxidizer streams, respectively. The definition of ν is given by

$$\nu=\frac{\nu_O W_O}{\nu_F W_F} \tag{13}$$

where W_i are the species molecular weights.

To create the thermochemical table, a number of one-dimensional flat flames at different equivalence ratios are computed with FlameMaster [38] using detailed chemistry, mixture-averaged molecular diffusion [39], and Soret diffusion. The reduced order model of Schlup and Blanquart [40] was used for the thermodiffusion coefficients. The necessary thermodynamic and transport properties, $(T,\rho,\mu,D_Z,D_Z^*,D_Z^T,D_C,\dot{\omega}_C)$, are interpolated onto a two-dimensional chemistry table as a function of C and Z. The 1D flamelets were calculated for equivalence ratios between $\phi=0.25$ and $\phi=1.3$. For more details on the full derivation of the new chemistry model, the reader is referred to [18]. The model has been validated across a wide range of configurations at DNS resolutions, including 1D, 2D, and 3D laminar freely propagating flames, 2D tubular flames, and 3D turbulent flames [19].

The filtered progress variable transport equation takes the same form as Eq. (5) with one difference. Now, $\bar{\dot{\omega}}_C$ is a function of both C and Z. Following the a priori analysis of Berger et al. [41], the filtered source term can be expressed as

$$\bar{\dot{\omega}}_C\left(\widetilde{C},C_v,\widetilde{Z}\right)=\iint\dot{\omega}_C(C,\phi)P\left(C|\widetilde{C},C_v\right)\delta\left(\phi-\hat{\phi}\right)dCd\phi \tag{14}$$

where δ is the delta distribution. To generate the table, one-dimensional flat flames at varying nominal equivalence ratios are filtered using a β-PDF in C. The results are tabulated in terms of $\widetilde{C}$, C_v, and $\widetilde{Z}$. Berger et al. [41] showed that using a β-PDF for the progress variable significantly reduced the source term error, and including higher order moments did not significantly increase the accuracy of the LES modelling.

It is important to note that the two-equation model is meant to capture the combustion dynamics both across the flame, parameterized by $\widetilde{C}$, and along the flame front, parameterized by $\widetilde{Z}$. Differential diffusion effects are strongly coupled to the local flame curvature, with regions of positive curvature having locally rich mixtures, and regions of negative curvature having locally lean mixtures (centre of curvature located in the burnt mixture). Thus, for the model to be most effective, the grid resolution should be sufficient to resolve smaller-scale structures on the order of the flame thickness.

5. LES results of the combustion chamber

The simulations of all combustion chambers are conducted on a cylindrical mesh of $N_x\times N_r\times N_\theta=256\times120\times256$ grid points. Along the inlet pipe and combustion chamber walls, the grid is refined in the radial direction. The grid is uniform in the axial direction until $x=0.08$ m before grid stretching is applied. This grid resolution is on the order of the laminar flame thickness at the flame front. The experimental data used for comparison are taken from Cheng et al. [1].

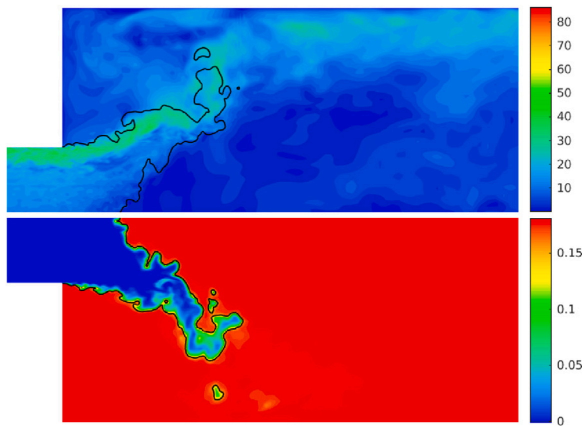


Fig. 2. Instantaneous profiles of velocity magnitude (top) and progress variable (bottom) for the simulation of the turbulent flame of methane/air. An isocontour at $C_{peak} = 0.149$ indicates the location of the flame front (black line).

5.1. Methane flame

The first experiment under consideration is a lean premixed methane/air flame ($\phi = 0.59$). Because methane is a unity Lewis number fuel, this simulation is conducted with Model 1 (see Section 4.1). It serves to validate the boundary conditions, grid resolution, and treatment of the tabulated chemistry within the LES framework. The progress variable is defined as the sum of the mass fractions of CO_2, CO, H_2O, and H_2.

Fig. 2 shows the instantaneous contour plots of the velocity magnitude and the progress variable. The flame location is marked with a black contour at C_{peak}, the location of the peak source term in progress variable space from the one-dimensional laminar flame. Because of the high axial velocity and high swirl, the flow at the edge of the inner pipe opens up (at an angle close to 45°) and creates a recirculation region slightly downstream of the inlet. This recirculation region is the key flow feature responsible for the stabilization of the flame front. The flame exhibits an "M" shape and is anchored at the edges of the inlet pipe. Because the combustion chamber is enclosed, hot gases are entrained in the recirculation regions in the corners and heat the outer shear layers that are associated with the inflow profile. Note that the present simulation framework does not include heat losses that are likely to occur at the burner exit, which may influence the visible luminosity of the flame.

The mean centreline velocity is shown in Fig. 3. The decay of the mean axial velocity is well reproduced by the simulations. More importantly, the location of the recirculation zone (represented by a negative mean axial velocity) is captured reasonably well. The start of the recirculation region is at 4.7 cm from the simulation and 6.6 cm from the experiments.

Three axial locations are chosen to investigate key regions of interest in the flow field. The station at $x = 0.015$ m is close to the inlet and serves to verify the inflow boundary conditions. The station at $x = 0.03$ m is chosen after the flow field starts to open, and as such, shows high sensitivity to the accuracy of the model. Finally, the station at $x = 0.08$ m is farther into the chamber, and represents a region where the flow is fully opened and has had time to develop. The first two stations occur before the recirculation region, and the last one is further downstream.

The axial and radial velocity profiles at these three axial locations are compared in Fig. 4. In all cases, the agreement is good. Some discrepancies in the mean quantities and fluctuations do exist, however. When considering these discrepancies, it is important to note that the experimental measurements suffer from some uncertainties. For instance, they are not perfectly symmetric and optical access to part

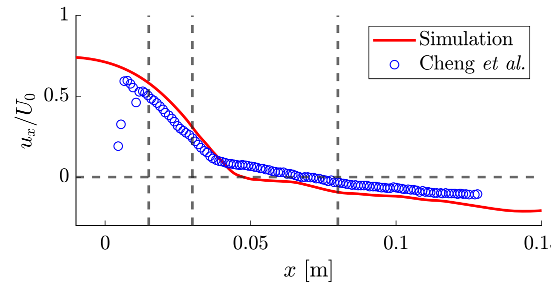


Fig. 3. Mean axial velocity along the centreline of the methane flame. The vertical dashed lines represent stations at three axial locations where radial statistics are collected.

of the domain was difficult (for $r < -30$ mm at $x = 80$ mm). The authors report contamination of the particle image velocimetry (PIV) data by laser reflection and particle deposition on the inner wall of the quartz enclosure [1]. Given these uncertainties, the agreement with experimental data is good, verifying the simulation framework.

5.2. Hydrogen flames

Cheng et al. [1] reported significant differences in the flow field between the methane and the hydrogen flames. In particular, they noted that the decay of the centreline velocity was lower, and there was larger acceleration in the post flame. This has the effect of pushing the inner recirculation region farther downstream, to $x \approx 0.13$ m.

The turbulent hydrogen flames are simulated using two different models, which are

- Model 1: $(\widetilde{C}, C_v)$ tabulation based on the one-dimensional laminar flamelet source term (see section 4.1),
- Model 2: $(\widetilde{C}, C_v, \widetilde{Z})$ tabulation based on the work of Schlup and Blanquart [18] (see section 4.2).

The progress variable fields for the two models are shown in Fig. 5. Differential diffusion has visible impacts on the flow field. There are two main observations. First, in model 1, the flame looks qualitatively similar to the methane flame. This result is expected since the only two parameters which differ between the two simulations are the density ratio between burnt and unburnt gases and the laminar flame speeds. The turbulent flames appear slightly shorter, a result of the higher laminar flame speed ($S_L = 0.206$ m/s for H_2 versus $S_L = 0.102$ m/s for CH_4). In contrast, model 2 exhibits a much shorter flame and is anchored closer to the burner entrance. Second, there are significant progress variable fluctuations in the burnt gases of model 2, whereas the burnt gases of model 1 are uniform. These fluctuations are attributed to Z capturing the curvature and differential diffusion effects.

The effect of the chemistry modelling on the flow field is illustrated in the mean velocity profiles shown in Figs. 6 and 7. In Fig. 6, the experimental data shows an increase in the mean axial velocity at the centreline after the flame, in clear contrast with the methane flame which shows a constant decay. This results in the recirculation region being pushed to $x \approx 0.13$ m. The $(\widetilde{C}, C_v)$ tabulation of model 1 completely fails to reproduce this unique flow feature, and the mean axial velocity decays just like that of the methane flame. Model 2 is able to reproduce the increase in axial velocity, its subsequent decay, and the ultimate location of the recirculation region. Although there are improvements in the flow field prediction, the experimentally measured velocity profile is smoother in the post flame region and does not exhibit the dip which is present in the simulation results. This could be due to underpredicted fluctuations in the x-direction. Additionally, Cheng et al. [1] reported significant broad-band noise generation from the hydrogen flame, which may play a role. However,

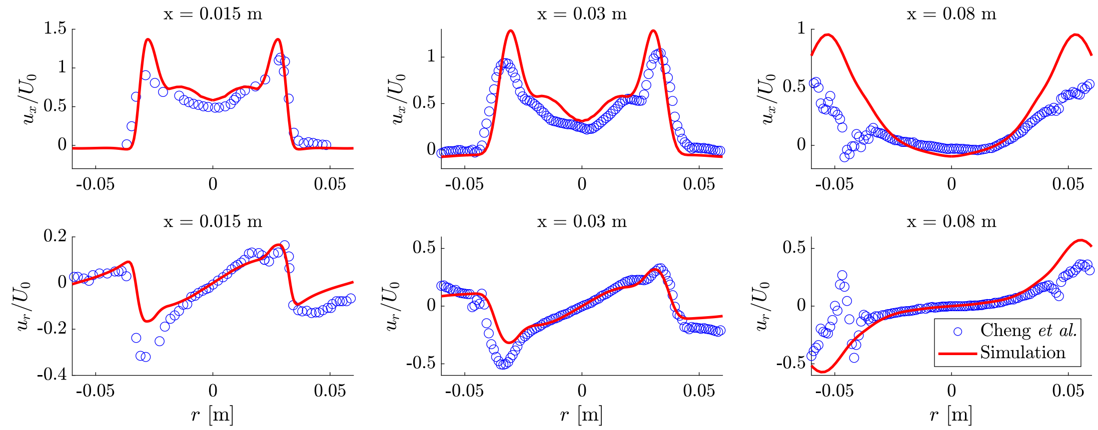
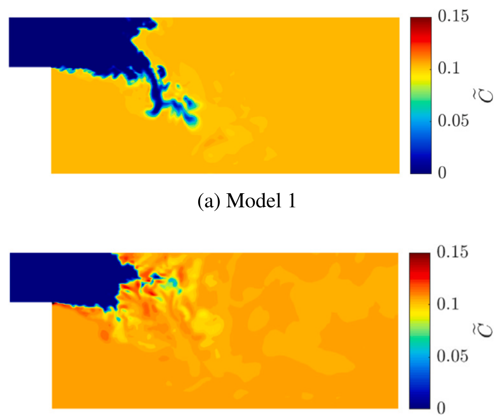


Fig. 4. Mean radial profiles of the axial (top) and radial (bottom) velocity components of the methane/air flame.



(a) Model 1

(b) Model 2

Fig. 5. Progress variable fields for the two different hydrogen tabulated chemistry models.

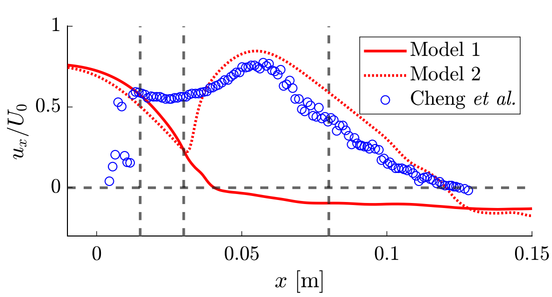


Fig. 6. Mean axial velocity along the centreline of the hydrogen flame comparing the two LES models.

acoustic effects are not captured by the present low Mach solver. Overall, the two-equation model, derived to capture the effects of curvature and differential diffusion, is able to better reproduce the unique flow features compared to the one-equation model.

The failure of the one-equation model is further illustrated in Fig. 7. At the first station, the profiles show that model 1 does not have the correct spreading rate, as the radial velocity is underestimated. At the second station, the radial velocity is still underpredicted for model 1, and slightly overpredicted for model 2. At the centreline, both models underpredict the axial velocity. At the final station, model 2 shows the increase in axial velocity at and around the centreline, correctly reproducing the push of the recirculation region downstream. Overall, the improvements in the axial and radial velocity profiles indicate that the modelling approaches in model 2 are able to overcome the deficiencies of the $(\widetilde{C}, C_v)$ tabulation of model 1.

To quantify the filtering of the LES, Fig. 8 shows the mean progress variable variance conditioned on $\widetilde{C}$ obtained from the simulations using model 2 compared to the variance obtained from filtering one-dimensional laminar flames. The one-dimensional flames are filtered with a Gaussian kernel at varying filter widths, expressed as a multiple of the laminar flame thickness, and the variance is computed as:

$$C_v = \widetilde{CC} - \widetilde{C}\widetilde{C} \tag{15}$$

For values of $\widetilde{C}$ close to 0 and C^{1D}_{max}, the variance predicted from the LES follows the profile of $\Delta = 2l_F$. At intermediate values of $\widetilde{C}$, the variance lies between $\Delta = 0.5l_F$ and $\Delta = l_F$. On average, the effective filter width that the flame experiences is on the order of the laminar flame thickness. Once again, values at $\widetilde{C} > C^{1D}_{max}$ correspond to superadiabatic burning, not present in one-dimensional flat flames. To measure the effect of the subfilter PDF modelling, the ratio of the source term sampled both with variance (β-PDF) and without variance (δ-PDF) is computed. For most of the values, the values sampled using the β-PDF are below those of the δ-PDF. Overall, the ratio is within the range of 0.5 to 1.1. More details are contained in the supplementary material.

Fig. 9 shows a close-up view of the mixture fraction and normalized source term for model 2. The nominal value of the mixture fraction for the given equivalence ratio is 0.0116. The $\widetilde{Z}$ field in Fig. 9 shows significant fluctuations along and behind the flame front. These fluctuations indicate significant differential diffusion effects since $\widetilde{Z}$ is meant to capture the effects of local fluctuations in the equivalence ratio. The effect on the flame can be seen in the plot of the normalized source term. There is a clear dependence on the flame geometry, where in regions of positive curvature (centre of curvature located in the burnt mixture), the source term is seen to be greater than the one-dimensional laminar maximum. Conversely, in regions of negative curvature, the source term is reduced, and the flame is even extinguished in some areas. The addition of the Z equation is necessary to capture these geometry-dependent effects.

6. Conclusions

Lean hydrogen premixed flames are subject to thermodiffusive instabilities that strongly influence local and global combustion properties. In this work, large eddy simulations were conducted on a low-swirl burner using the tabulated chemistry framework. The simulations were conducted in two parts. First, a simulation of the low-swirl injector was performed using the immersed boundary method. This simulation was used to generate inflow boundary conditions for a second calculation of the combustion chamber.

The tabulated chemistry method was first validated on a methane/air flame at an equivalence ratio of $\phi = 0.59$, since it exhibits unity

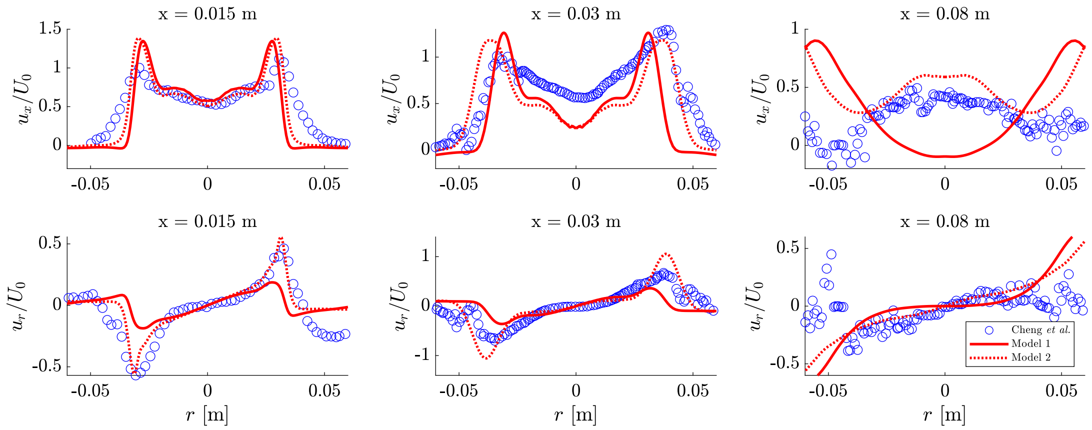


Fig. 7. Mean radial profiles of the axial (top) and radial (bottom) velocity components of the hydrogen/air flame.

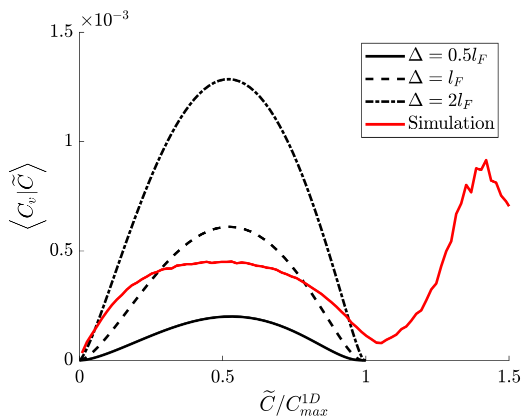


Fig. 8. Conditional mean variance from the simulation (model 2) compared to variance obtained from filtering one-dimensional flames at varying filter widths Δ.

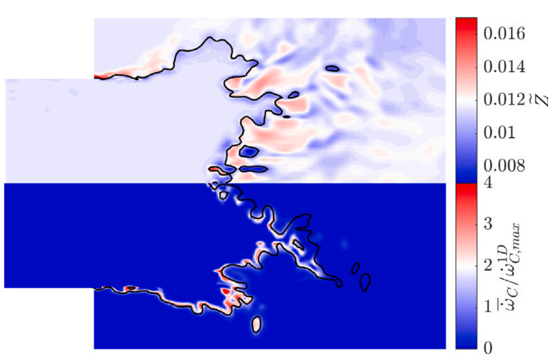


Fig. 9. Close-up view of the mixture fraction (top) and normalized source term (bottom) for model 2. The black line is an isocontour at C_{peak} = 0.0853 to indicate the flame location.

Lewis number behaviour. The flame opens up at about a 45 degree angle and is stabilized by the presence of a recirculation region. Then, LES were conducted on a premixed hydrogen/air flame at an equivalence ratio of $\phi = 0.4$. Due to differential diffusion effects, the hydrogen flame exhibits a unique flow field in the experiments. In particular, the recirculation region is pushed further downstream due to larger acceleration of the flow behind the flame.

Two filtered tabulation models were tested in the LES. The first is the classical one-equation tabulation, similar to the case of the methane flame. This method was unable to reproduce correct mean statistics in terms of axial or radial velocity profiles. The second method was based on a two-equation tabulation, recently proposed by Schlup and Blanquart [18]. This model was able to capture the differential diffusion effects and reproduced successfully the unique flow field of the turbulent hydrogen flame. This work is the first successful application of the two-equation model in the LES framework.

Novelty and Significance Statement

The novelties of this method are two-fold: 1) the extension of a recently proposed tabulated chemistry model which captures differential diffusion effects to large eddy simulation and 2) its validation in the simulation of a low swirl combustor. This is significant because it marks a step forward in developing LES models for lean hydrogen turbulent premixed flames, which are notoriously difficult to model due to their propensity for thermodiffusive instabilities. The method successfully reproduces the unique flow field exhibited by hydrogen flames in the combustor.

CRediT authorship contribution statement

Matthew X. Yao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Visualization. **Guillaume Blanquart:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.proci.2024.105500. The supplementary material contains details regarding the immersed boundary implementation and the effect of the subfilter PDF on the source term.

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